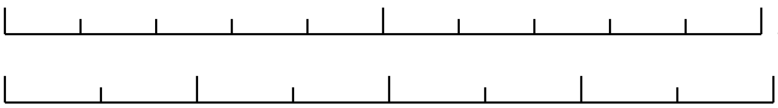
Powder doser — PCB 1:1 print kit (color)

Quilter run-3 candidate 2 with the In2 GND-plane fix — the board prepared for JLCPCB ordering (PR #76, note 25). Board: 110 x 110 mm, 4-layer (F.Cu / In1 GND / In2 GND / B.Cu), 16 through-hole footprints. Generated 2026-07-16.

How to print

1. Print in COLOR on US Letter at 100% / “Actual size” — turn OFF “fit to page” / “shrink to fit”.
2. Check the calibration rulers below with a real ruler; every board page also carries a 100 mm bar.
3. Page 2 (top assembly overlay) is the one to lay the physical parts on: grey outlines are the real part bodies, gold shapes are the pad/hole openings, purple is the courtyard.
4. Bottom/inner-layer pages are X-ray views (not mirrored), so they align with page 2 held to the light.

Calibration — these must measure exactly 100 mm and 4 in:



100 mm

4 in

Parts to gather for the spot check

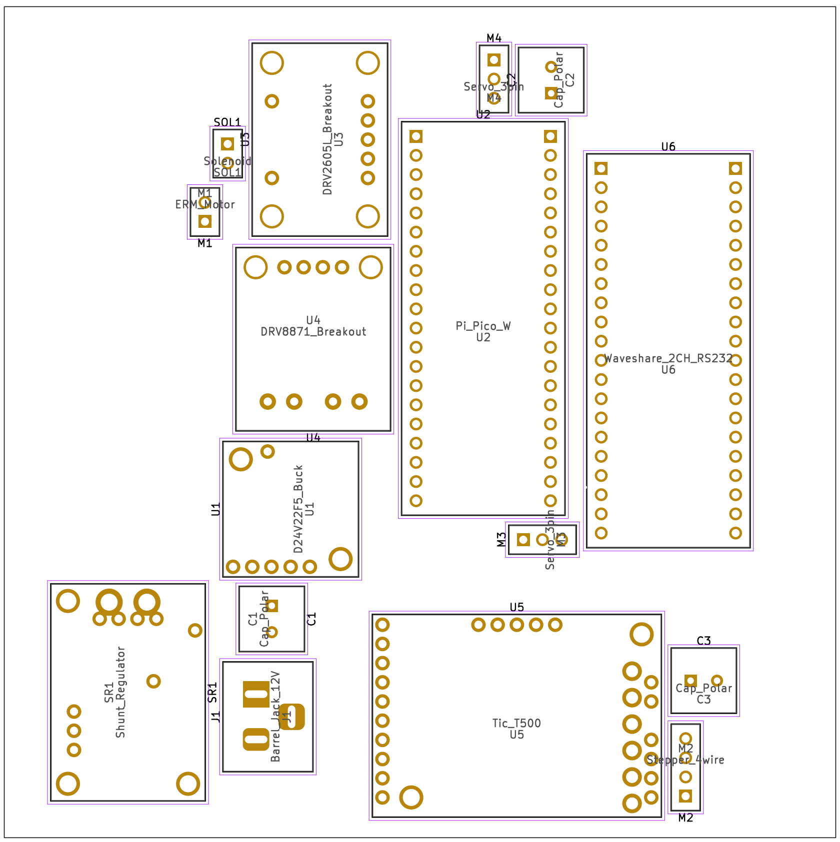
Ref	Part	What it looks like
U1	Pololu D24V22F5 5 V buck	17.8 x 17.8 mm carrier, 5+1 holes
U2	Raspberry Pi Pico W	2x20 pins @ 17.78 mm row spacing
U3	Adafruit DRV2605L breakout (STEMMA QT)	haptic driver, 25.4 x 17.78 mm
U4	Adafruit DRV8871 breakout	solenoid driver, 2x 3.5 mm terminal blocks
U5	Pololu Tic T500	stepper controller, 38.1 x 26.7 mm
U6	Waveshare Pico-2CH-RS232 (scale)	2x20 side receptacle, 21 x 52 mm body
SR1	Pololu shunt regulator (#3776)	28.6 x 20.3 mm, VIN/GND terminals
J1	Barrel jack 12 V (PJ-102AH)	tip=+12V, sleeve=GND
C1	100 uF radial cap (buck VIN)	D8 mm, 3.5 mm pitch, square pad = +
C2	100 uF radial cap (Pico VSYS)	D8 mm, 3.5 mm pitch, square pad = +
C3	100 uF radial cap (Tic VIN)	D8 mm, 3.5 mm pitch, square pad = +
M1	ERM vibration motor header	1x2 pin header 0.1 inch
M2	Stepper (NEMA-11) header	1x4 pin header 0.1 inch
M3	Servo 1 header (SERVO_SIG)	1x3 pin header 0.1 inch
M4	Servo 2 header (SERVO_SIG2)	1x3 pin header 0.1 inch
SOL1	Solenoid header	1x2 pin header 0.1 inch

Contents: 1 cover · 5 board pages (1:1) · drill map + schematic (reference, not to scale). Source files in PR #76, paper/background/starter_board/quilter_candidates_run3/ — rebuild with: python3 make_print_pdf.py

Top assembly overlay — lay the parts on this page

SCALE 1:1 — print at 100% / “Actual size”

Grey = part body (F.Fab), black = silkscreen, purple = courtyard, gold = pad/hole openings (F.Mask).

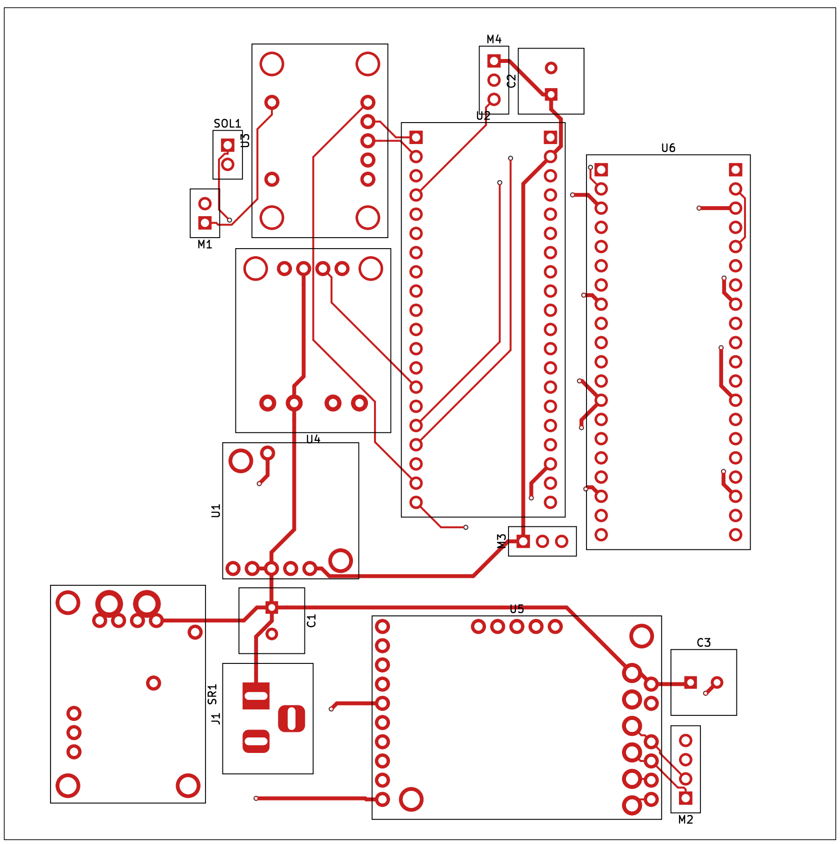


0 100 mm (verify with a ruler)

Top copper (F.Cu) + silkscreen

SCALE 1:1 — print at 100% / “Actual size”

Red = top copper as routed by Quilter (run 3, candidate 2).

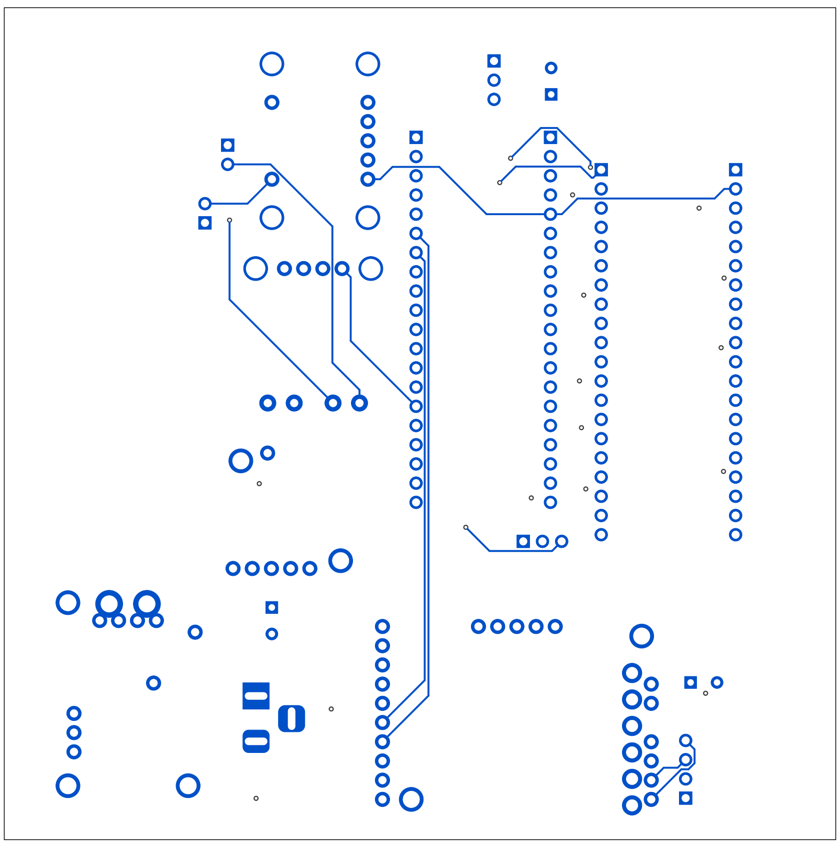


0 100 mm (verify with a ruler)

Bottom copper (B.Cu) — X-ray view

SCALE 1:1 — print at 100% / “Actual size”

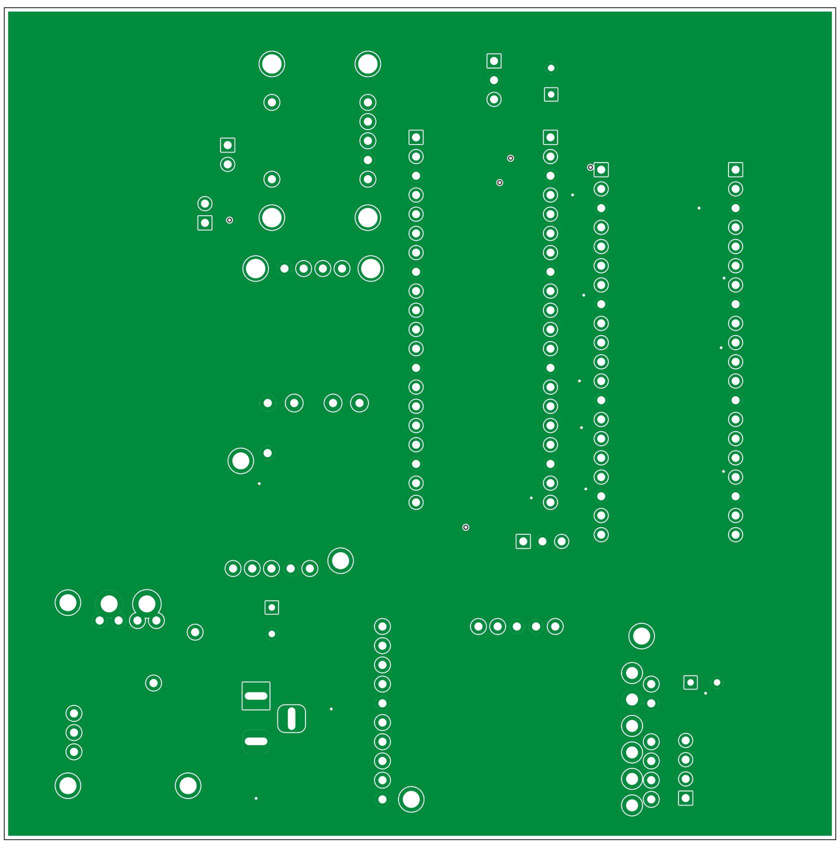
Blue = bottom copper, NOT mirrored (drawn looking through the board), so it overlays the top pages when held to the light.



Inner layer 1 (In1.Cu) — GND plane

SCALE 1:1 — print at 100% / “Actual size”

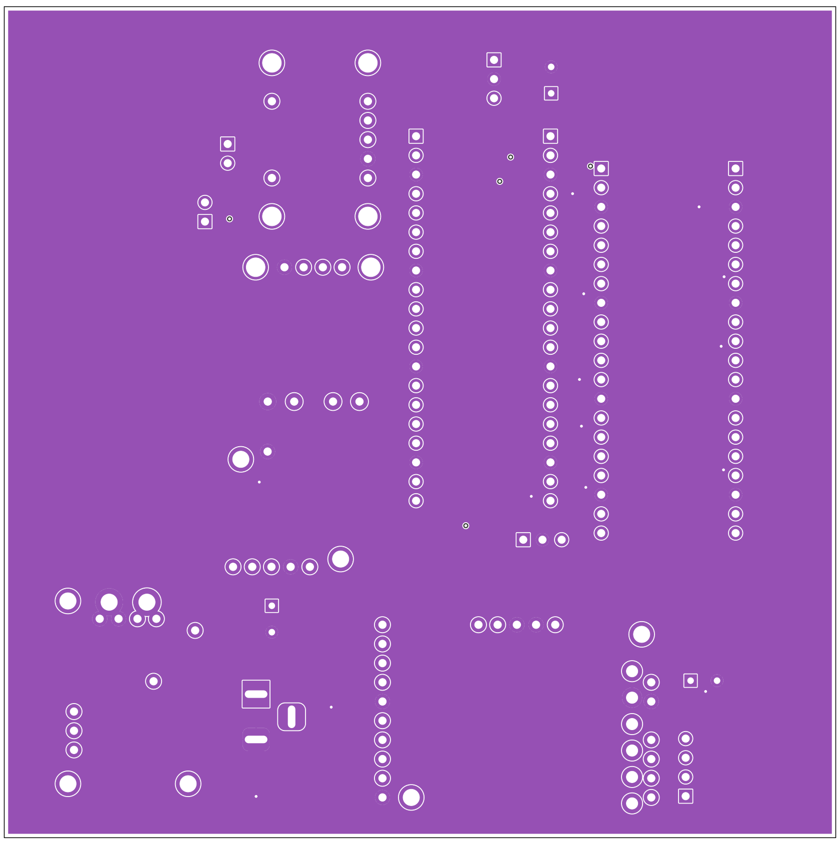
Green = inner-layer copper (solid GND plane); X-ray view.



Inner layer 2 (In2.Cu) — GND plane

SCALE 1:1 — print at 100% / “Actual size”

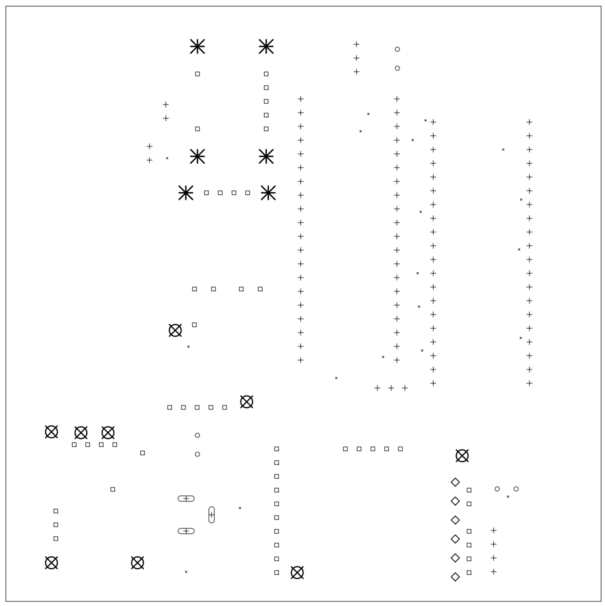
Purple = inner-layer copper (former <no net> zone, now GND per the note-25 fix); X-ray view.



0 100 mm (verify with a ruler)

Drill map — hole symbols and size table

Plated holes for all 16 THT parts; see the legend table for sizes.



Drill Map:

- · 0.305mm / 0.0120'' (19 holes)
- ○ 0.800mm / 0.0315'' (6 holes)
- + 1.000mm / 0.0394'' (94 holes + 3 slots)
- □ 1.020mm / 0.0402'' (51 holes)
- ◇ 1.520mm / 0.0598'' (6 holes)
- ⊠ 2.180mm / 0.0858'' (9 holes)
- ✳ 2.500mm / 0.0984'' (6 holes)

Schematic — test_module_unplaced.kicad_sch (2-servo, rev C nets)

NOT TO SCALE — reference only

23 nets / 99 connected pins; matches the board netlist exactly.